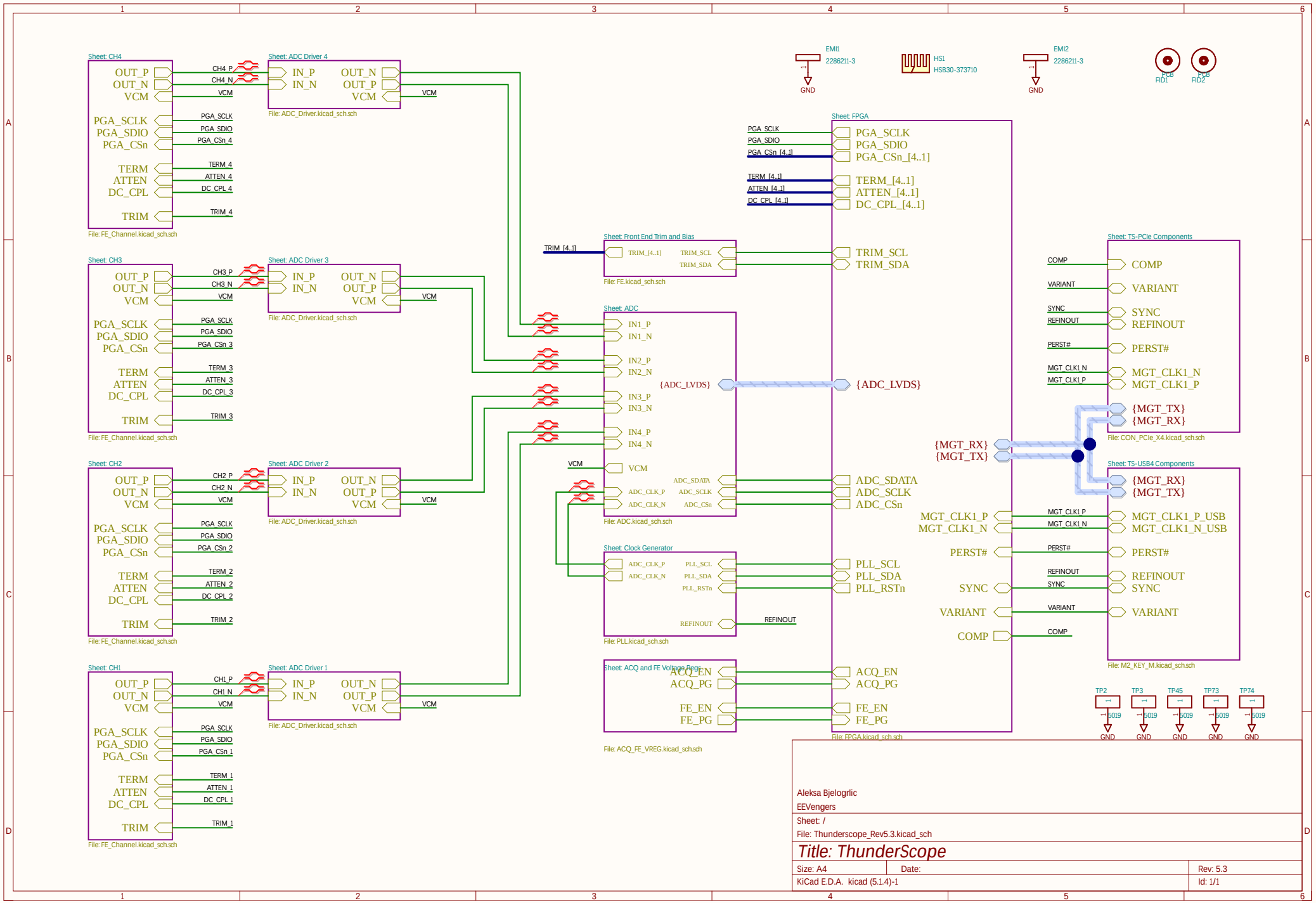
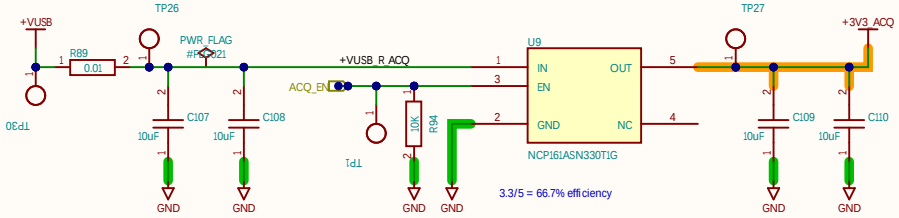
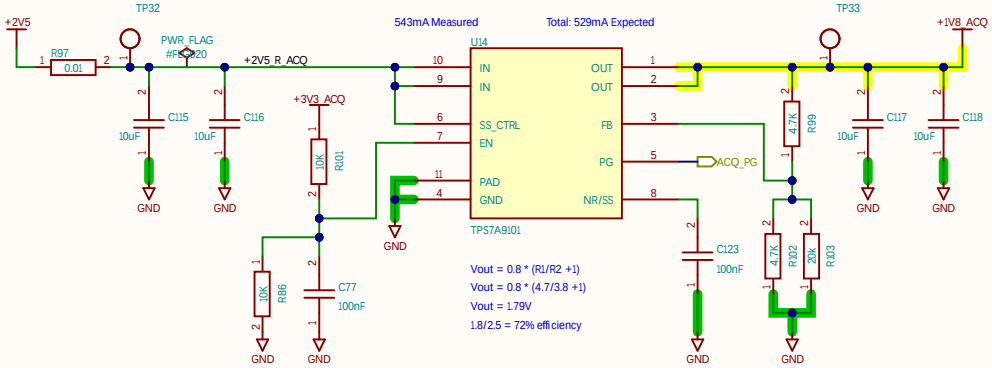
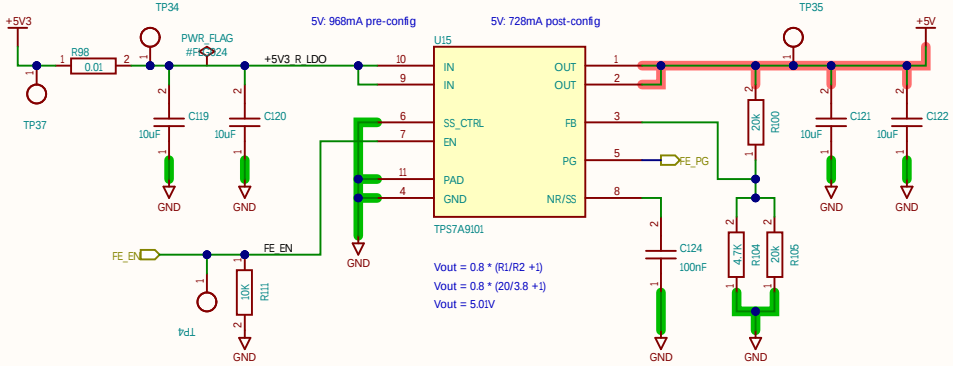

Sheet: CH4
OUT_P
OUT_N
VCM
PGA_SCLK
PGA_SDIO
PGA_CSn
TERM
ATTEN
DC_CPL
TRIM
CH4_P
CH4_N
PGA_SCLK
PGA_SDIO
PGA_CSn_4
TERM_4
ATTEN_4
DC_CPL_4
TRIM_4
File: FE_Channel.kicad_sch.sch
Sheet: ADC Driver 4
IN_P
IN_N
OUT_N
OUT_P
VCM
File: ADC_Driver.kicad_sch.sch
Sheet: CH3
CH3_P
CH3_N
PGA_CSn_3
TERM_3
ATTEN_3
DC_CPL_3
TRIM_3
Sheet: ADC Driver 3
Sheet: CH2
CH2_P
CH2_N
PGA_CSn_2
TERM_2
ATTEN_2
DC_CPL_2
TRIM_2
Sheet: ADC Driver 2
Sheet: CH1
CH1_P
CH1_N
PGA_CSn_1
TERM_1
ATTEN_1
DC_CPL_1
TRIM_1
Sheet: ADC Driver 1
EMI1
2286211-3
GND
HS1
HSB30-373710
EMI2
FCB
FID1
FID2
Sheet: FPGA
PGA_CSn_[4..1]
TERM_[4..1]
ATTEN_[4..1]
DC_CPL_[4..1]
TRIM_SCL
TRIM_SDA
{ADC_LVDS}
ADC_SDATA
ADC_SCLK
ADC_CSn
PLL_SCL
PLL_SDA
PLL_RSTn
ACQ_EN
ACQ_PG
FE_EN
FE_PG
{MGT_RX}
{MGT_TX}
MGT_CLK1_P
MGT_CLK1_N
PERST#
SYNC
VARIANT
COMP
File: FPGA.kicad_sch.sch
Sheet: Front End Trim and Bias
TRIM_[4..1]
File: FE.kicad_sch.sch
Sheet: ADC
IN1_P
IN1_N
IN2_P
IN2_N
IN3_P
IN3_N
IN4_P
IN4_N
ADC_CLK_P
ADC_CLK_N
File: ADC.kicad_sch.sch
Sheet: Clock Generator
REFINOUT
File: PLL.kicad_sch.sch
Sheet: ACQ and FE Voltage Regs
File: ACQ_FE_VREG.kicad_sch.sch
Sheet: TS-PCIe Components
REFINOUT
MGT_CLK1_N
MGT_CLK1_P
File: CON_PCIe_X4.kicad_sch.sch
Sheet: TS-USB4 Components
MGT_CLK1_P_USB
MGT_CLK1_N_USB
File: M2_KEY_M.kicad_sch.sch
TP2
TP3
TP45
TP73
TP74
5019
GND
Aleksa Bjelogrlic
EEVengers
Sheet: /
File: Thunderscope_Rev5.3.kicad_sch
Title: ThunderScope
Size: A4
Date:
Rev: 5.3
KiCad E.D.A. kicad (5.1.4)-1
Id: 1/1

Acquisition Voltage Regulators

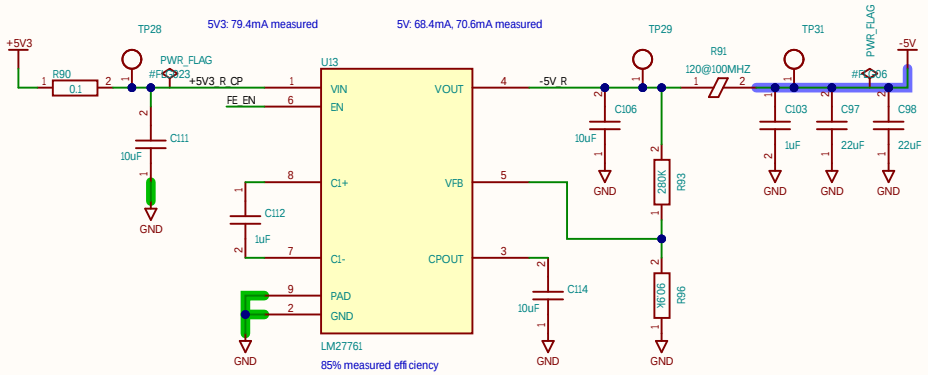


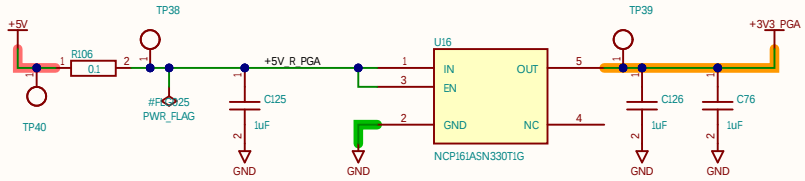


Front End Voltage Regulators

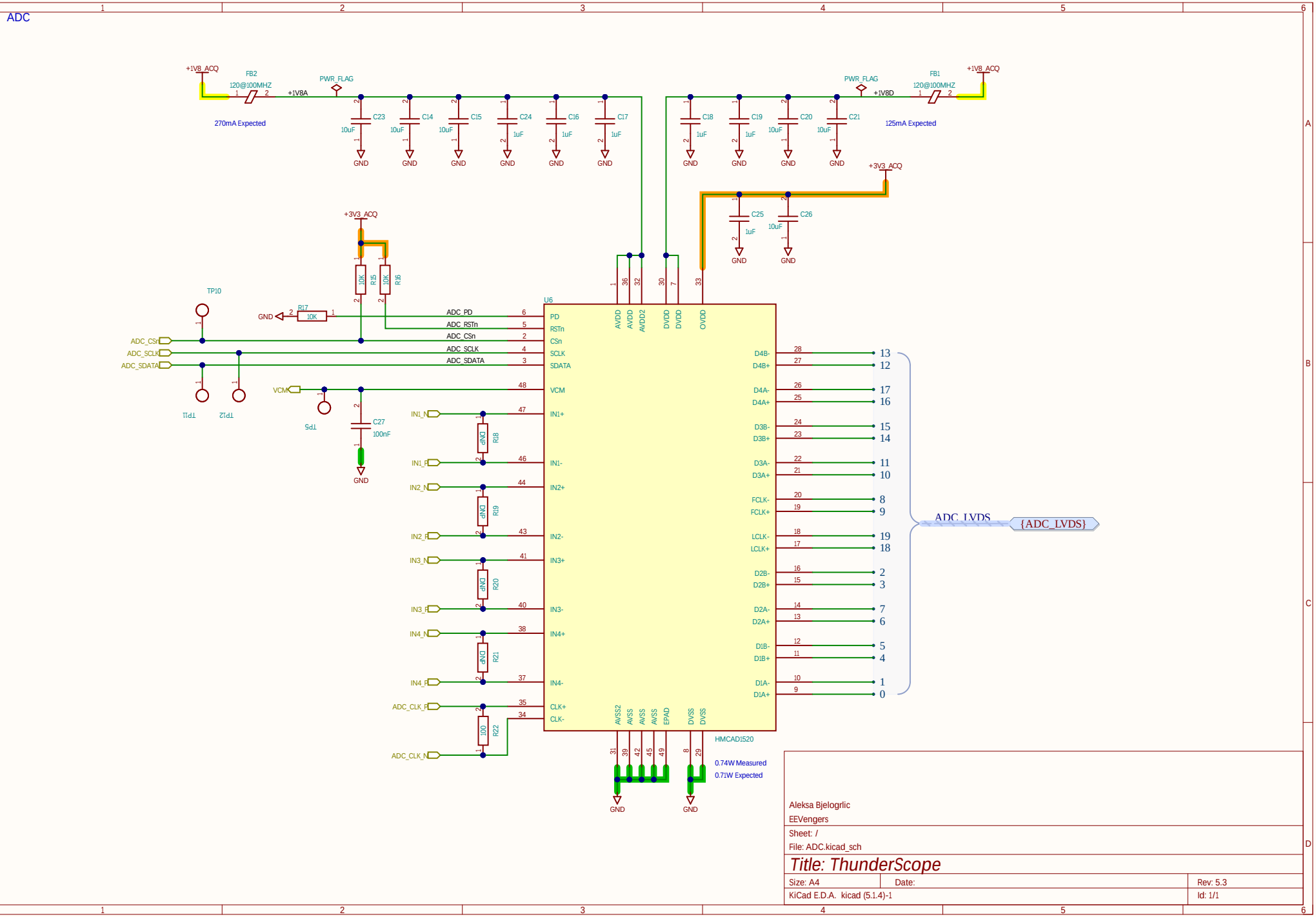


6ms soft start time on previous LDO, need same or greater
Tss = (VREF × Cnr/ss) / Inr/ss
Tss = (0.8 × 100nF) / 6.2uA [for SS_CTRL = GND]
Tss = 0.0129s = 13ms

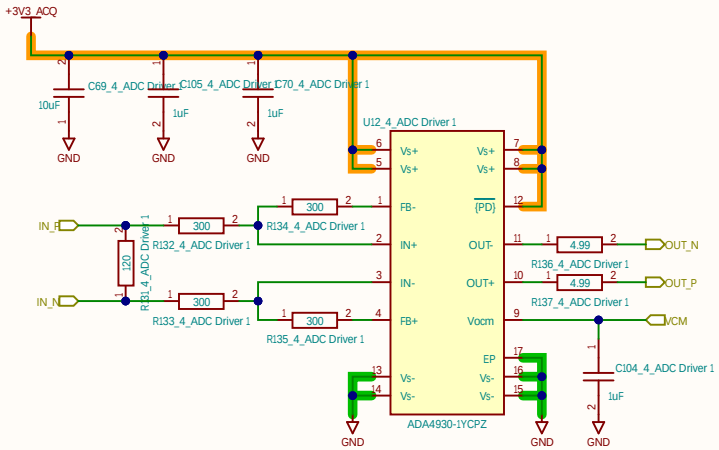




Aleksa Bjelogrlic		
EEVengers		
Sheet: /		
File: ACQ_FE_VREG.kicad_sch		
Title: ThunderScope		
Size: A4	Date:	Rev: 5.3
KiCad E.D.A. kicad (5.1.4)-1		Id: 1/1

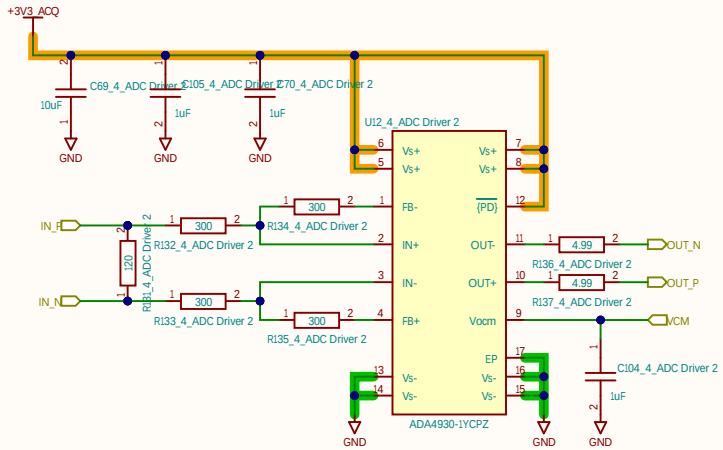

ADC
+1V8_ACQ
FB2
120@100MHZ
PWR_FLAG
+1V8A
270mA Expected
C23
C14
C15
C24
C16
C17
10uF
1uF
GND
+1V8D
FB1
C18
C19
C20
C21
125mA Expected
+3V3_ACQ
C25
C26
TP10
R17
10K
R15
R16
ADC_CSn
ADC_SCLK
ADC_SDATA
VCM
TP11
TP12
TP5
C27
100nF
U6
PD
RSTn
CSn
SCLK
SDATA
VCM
IN1+
IN1-
IN2+
IN2-
IN3+
IN3-
IN4+
IN4-
CLK+
CLK-
AVDD
AVDD2
DVDD
OVDD
AVSS2
AVSS
EPAD
DVSS
D4B-
D4B+
D4A-
D4A+
D3B-
D3B+
D3A-
D3A+
FCLK-
FCLK+
LCLK-
LCLK+
D2B-
D2B+
D2A-
D2A+
D1B-
D1B+
D1A-
D1A+
ADC_LVDS
{ADC_LVDS}
HMCAD1520
0.74W Measured
0.71W Expected
Aleksa Bjelogrlic
EEVengers
Sheet: /
File: ADC.kicad_sch
Title: ThunderScope
Size: A4
Date:
Rev: 5.3
KiCad E.D.A. kicad (5.1.4)-1
Id: 1/1

ADC Driver



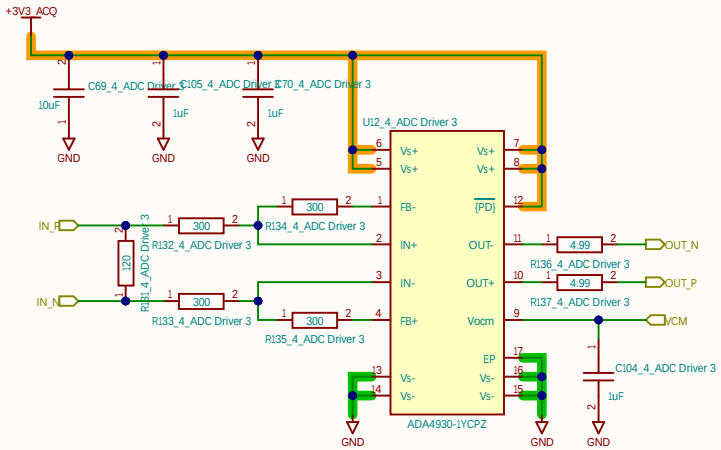
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EEVengers		
Sheet: /		
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Title: ThunderScope		
Size: A4	Date:	Rev: 5.3
KiCad E.D.A. kicad (5.1.4)-1		Id: 1/1

ADC Driver



Aleksa Bjelogrlic		
EEVengers		
Sheet: /		
File: ADC_Driver.kicad_sch		
Title: ThunderScope		
Size: A4	Date:	Rev: 5.3
KiCad E.D.A. kicad (5.1.4)-1		Id: 1/1

ADC Driver

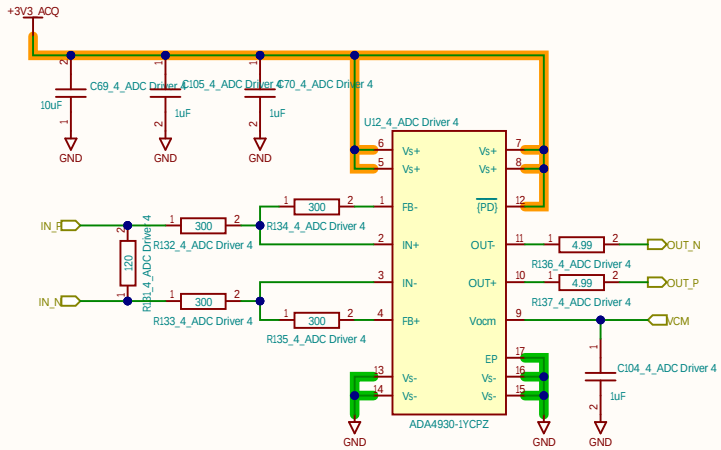


Aleksa Bjelogrlic
EEVengers
Sheet: /
File: ADC_Driver.kicad_sch

Title: ThunderScope

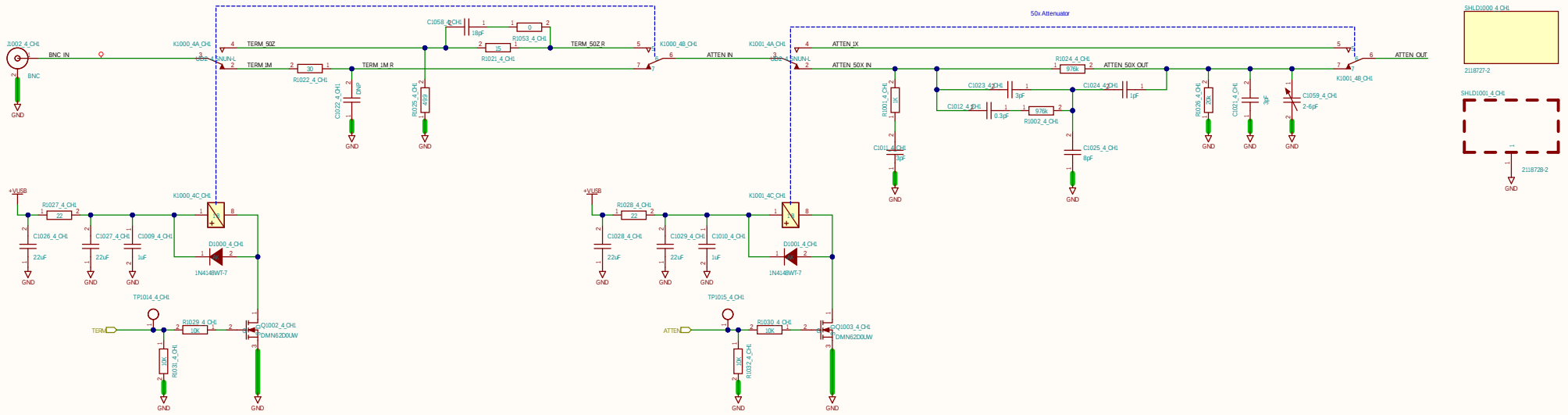
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KiCad E.D.A. kicad (5.1.4)-1		Id: 1/1

ADC Driver

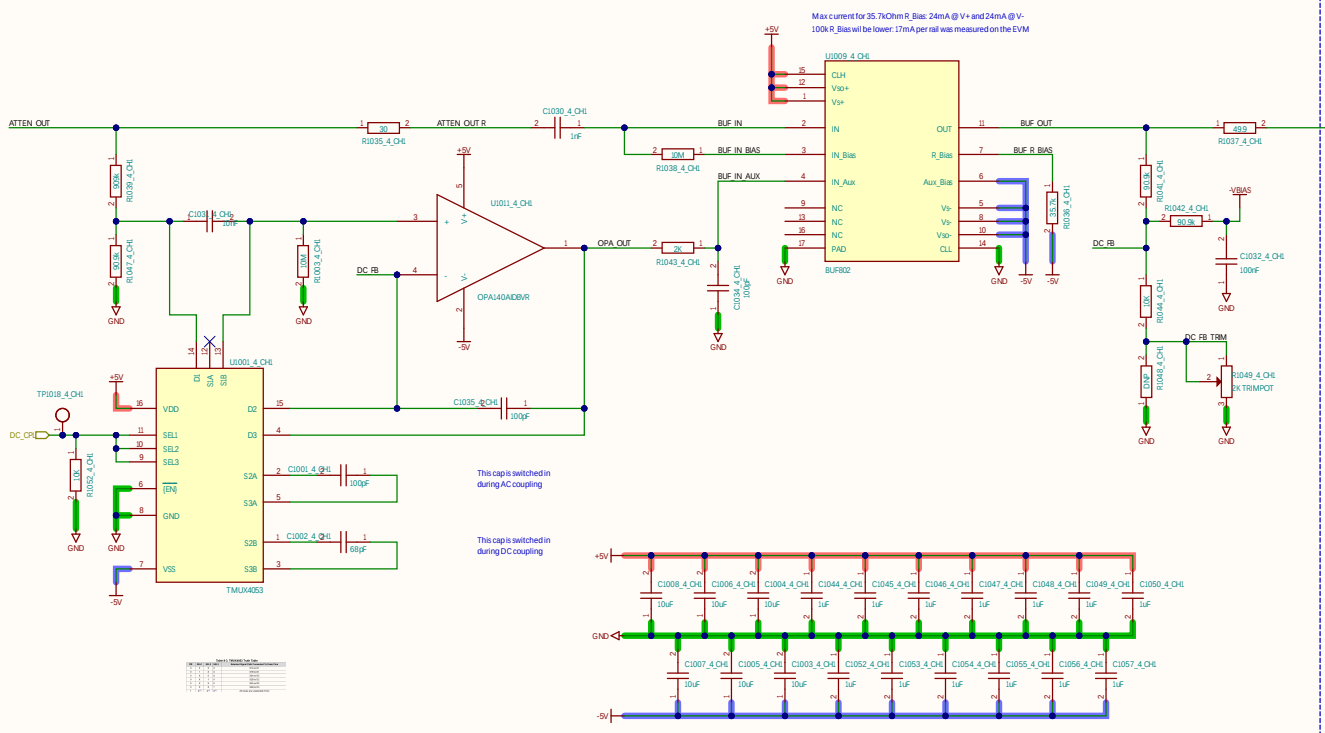


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KiCad E.D.A. kicad (5.1.4)-1		Id: 1/1

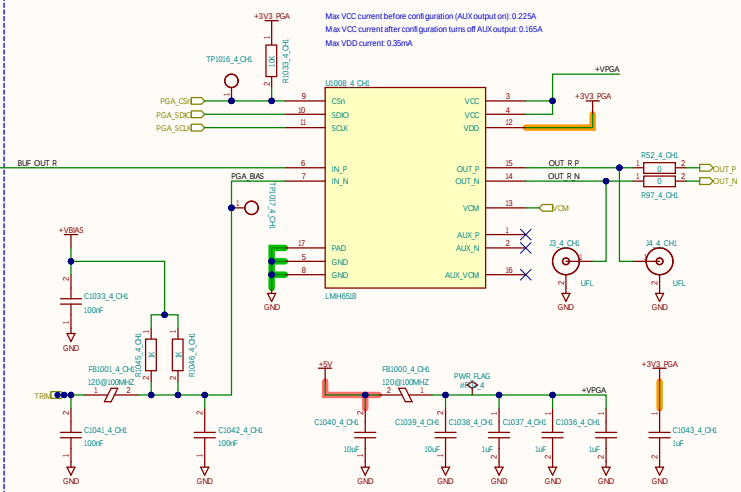
Termination and Attenuation



Input Buffer and AC/DC Coupling

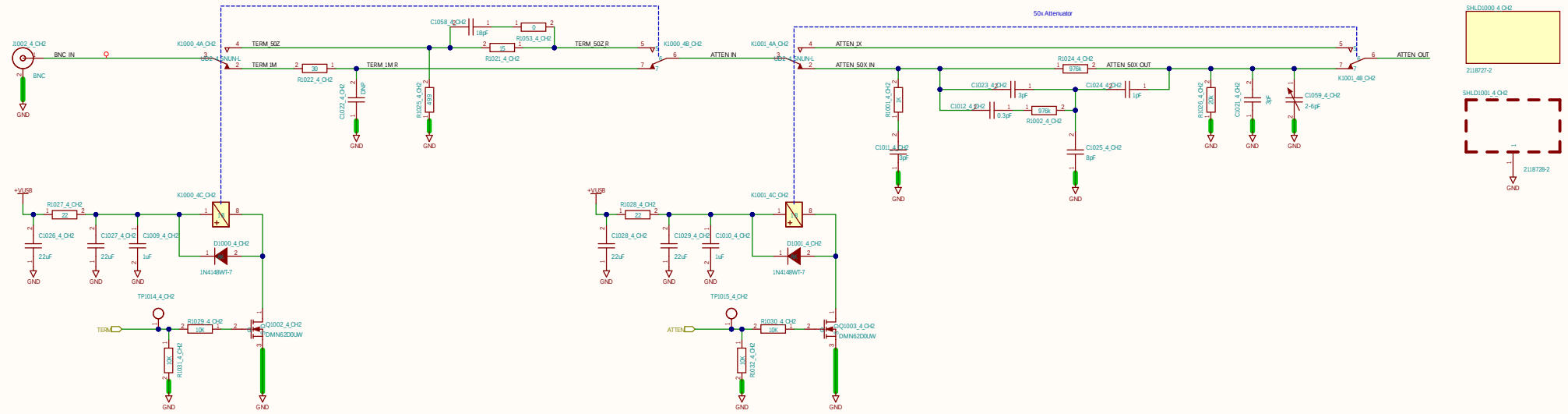


Programmable Gain Amplifier

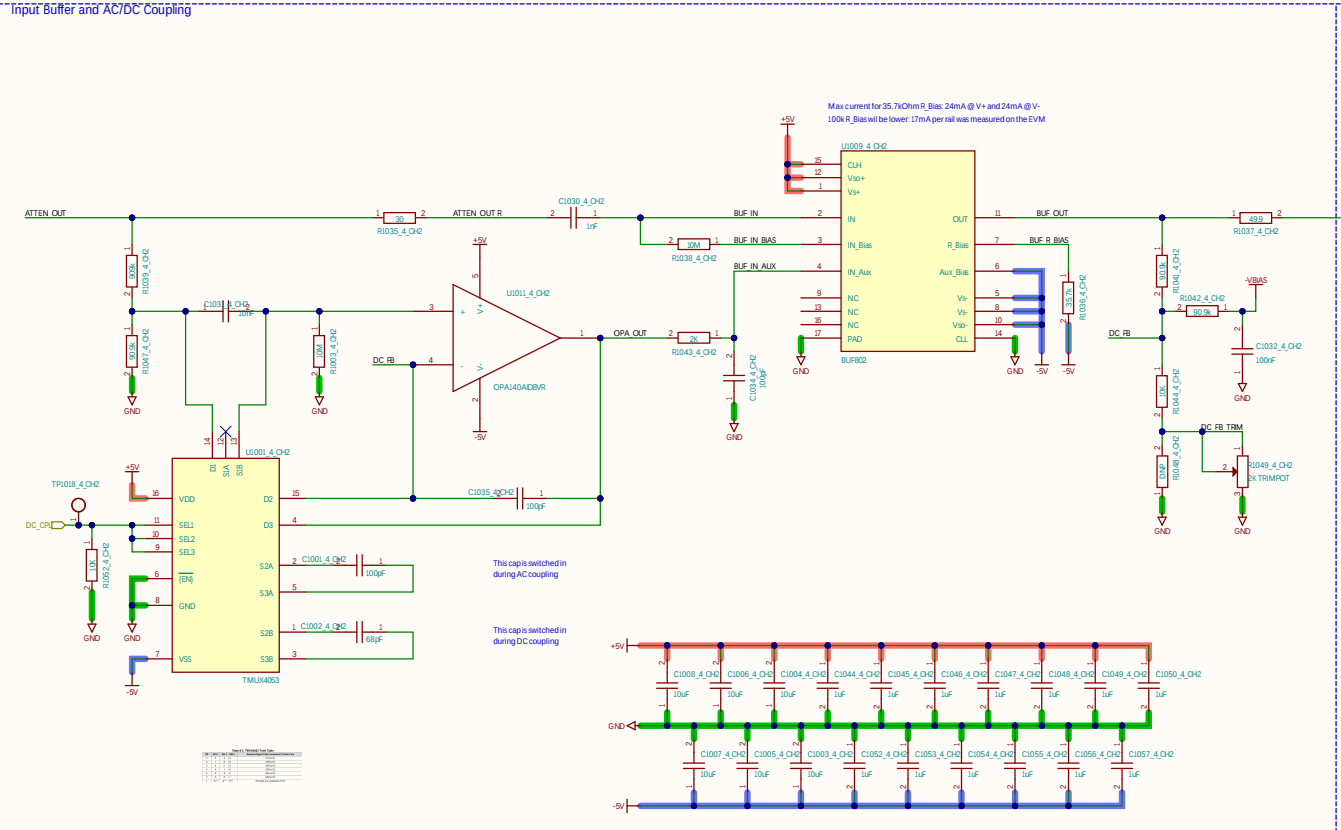


Aleksa Bjelogrlic	
EEVengers	
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Title: ThunderScope	
Size: A3	Date: Rev: 5.3
KiCad E.D.A. kicad (5.1.4) -1	Id: 1/1

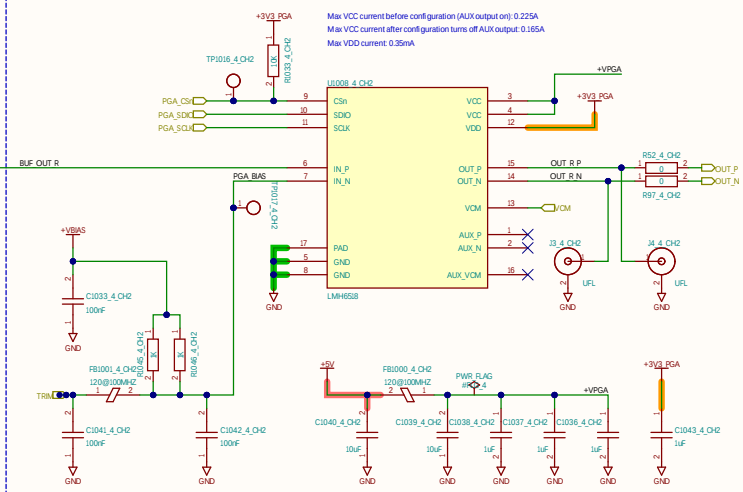
Termination and Attenuation



Input Buffer and AC/DC Coupling

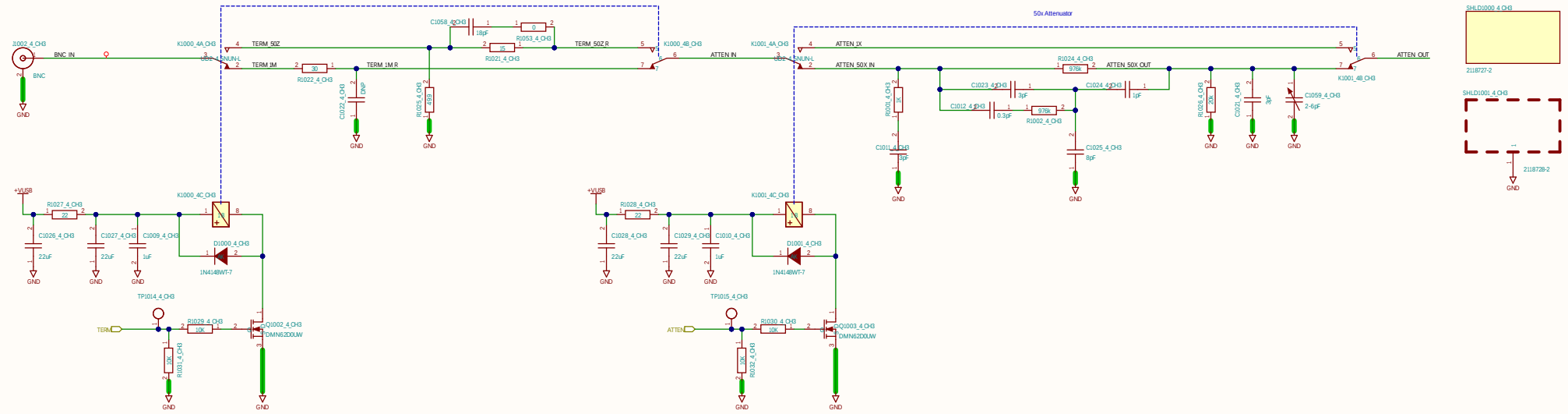


Programmable Gain Amplifier

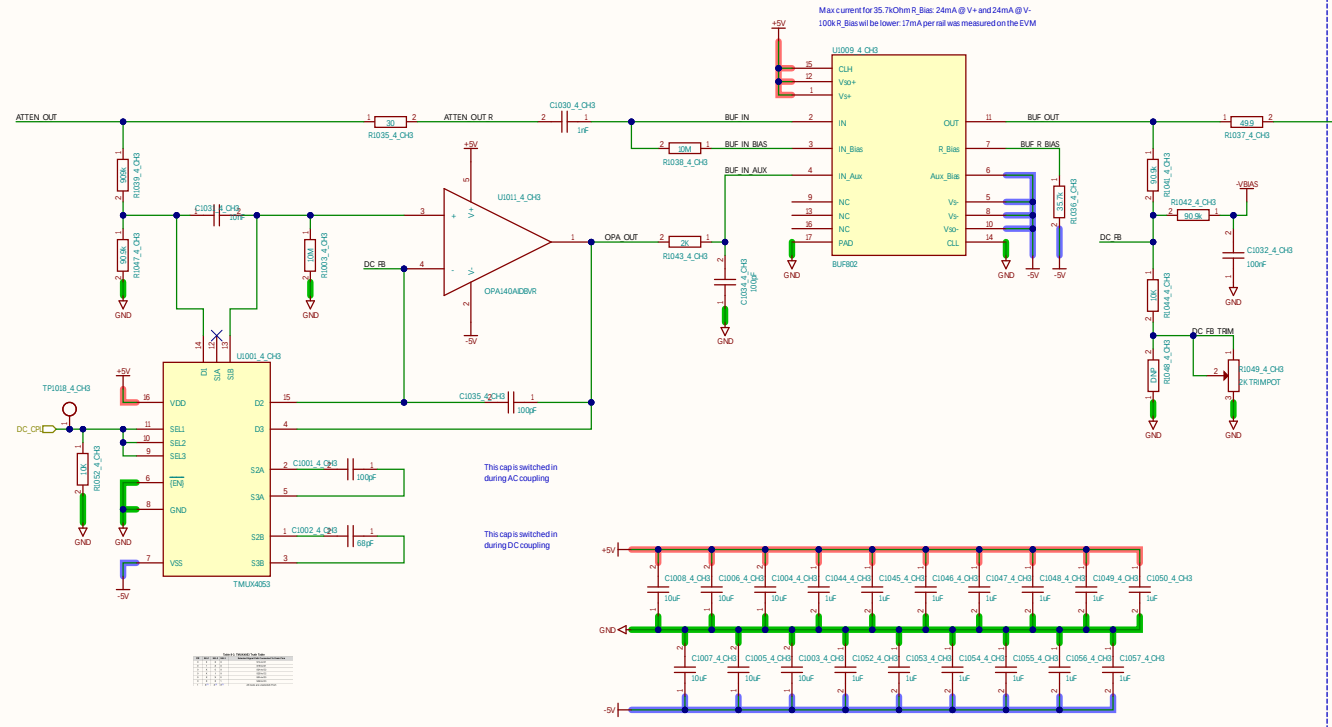


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EEVengers	
Sheet: /	
File: FE_Channel.kicad_sch	
Title: ThunderScope	
Size: A3	Date: Rev: 5.3
KiCad E.D.A. kicad (5.1.4) -1	Id: 1/1

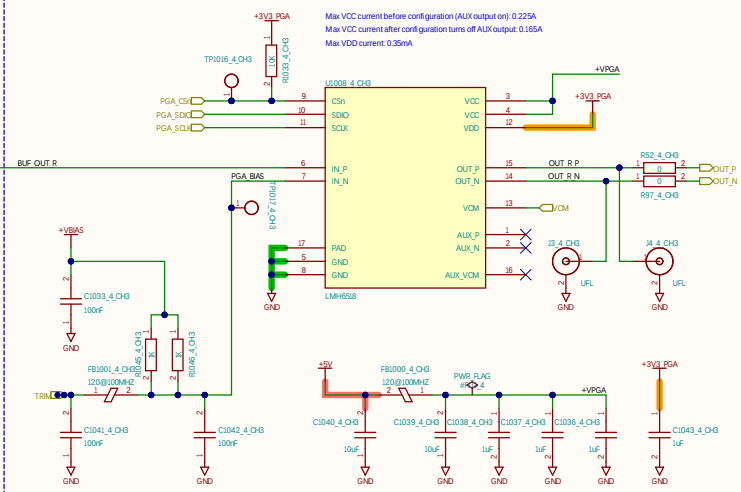
Termination and Attenuation



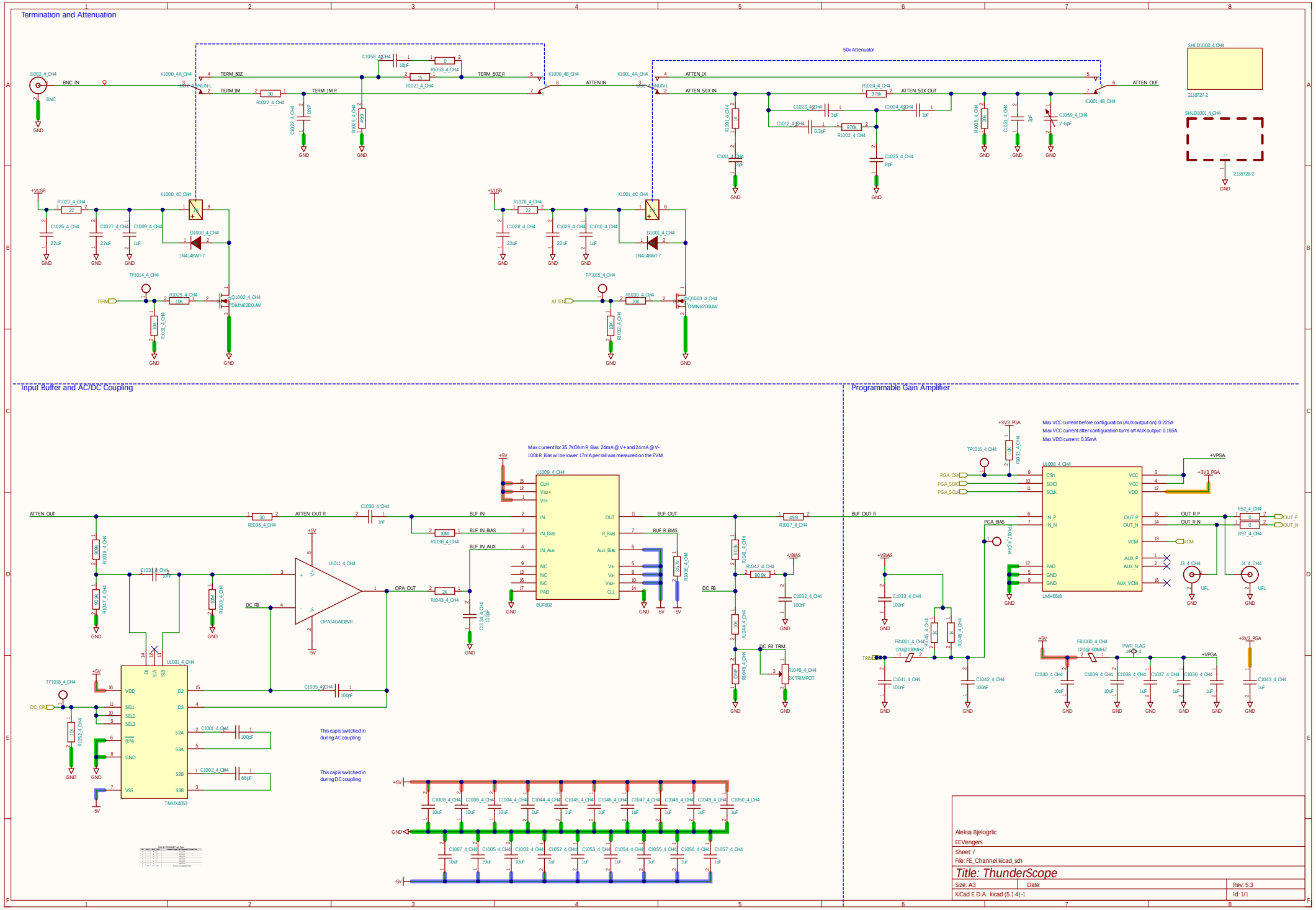
Input Buffer and AC/DC Coupling



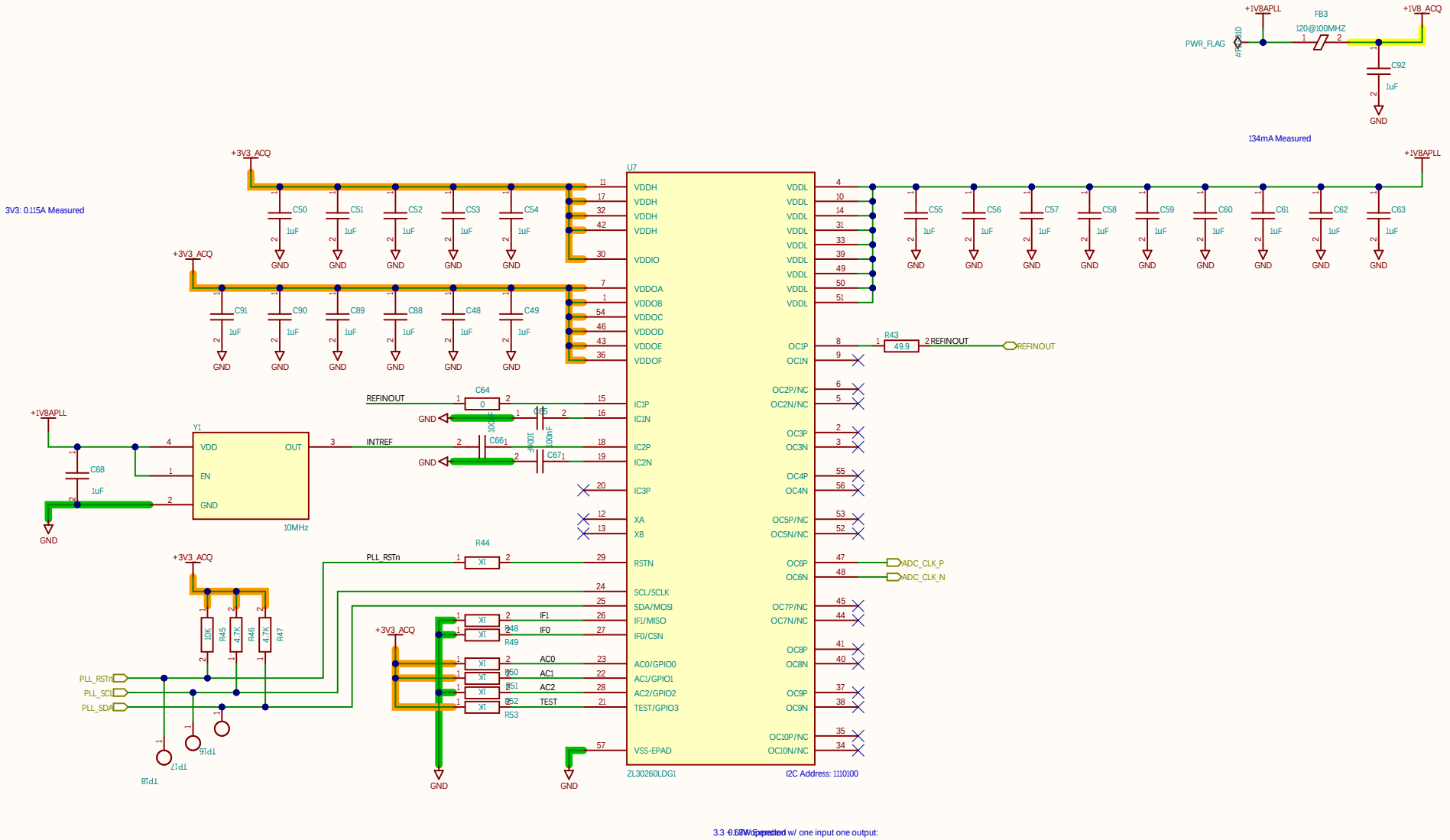
Programmable Gain Amplifier



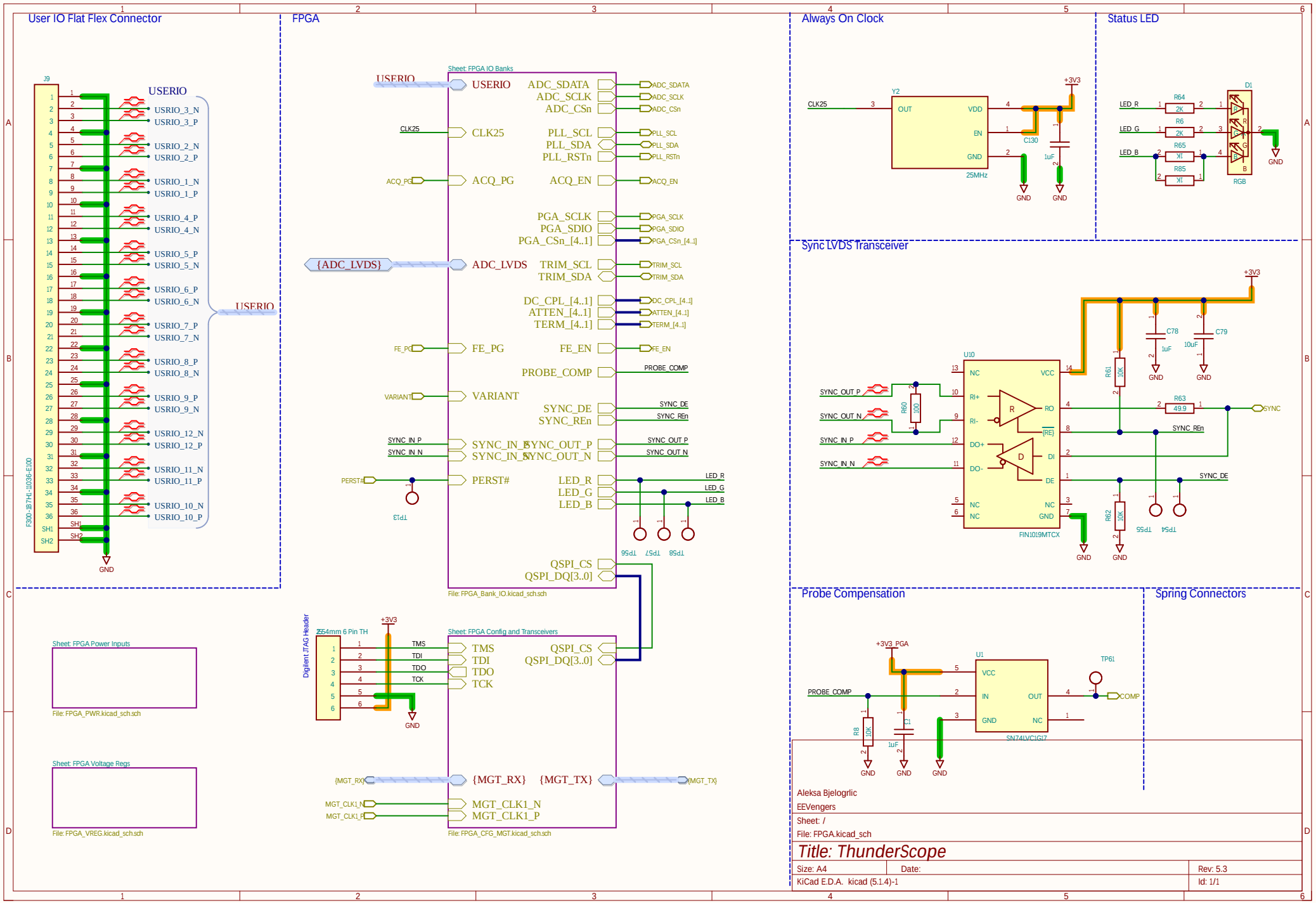
Aleksa Bjelogrlic	
EEVengers	
Sheet: /	
File: FE_Channel.kicad_sch	
Title: ThunderScope	
Size: A3	Date: Rev: 5.3
KiCad E.D.A. kicad (5.1.4) -1	Id: 1/1


Termination and Attenuation
50x Attenuator
BNC IN
TERM_50Z
TERM_1M
TERM_1M R
TERM_50Z R
ATTEN IN
ATTEN_1X
ATTEN_50X IN
ATTEN_50X OUT
ATTEN OUT
SHLD1000_4_CH4
2118727-2
SHLD1001_4_CH4
2118728-2
+VUSB
TERM
ATTEN
1N4148WT-7
DMN62D0UW
Input Buffer and AC/DC Coupling
ATTEN_OUT
ATTEN_OUT R
BUF_IN
BUF_IN_BIAS
BUF_IN_AUX
BUF_OUT
BUF_R_BIAS
OPA_OUT
DC_FB
DC_CPL
BUF802
OPA140AIDBVR
TMUX4053
Max current for 35.7kOhm R_Bias: 24mA @ V+ and 24mA @ V-
100k R_Bias wil be lower: 17mA per rail was measured on the EVM
This cap is switched in during AC coupling
This cap is switched in during DC coupling
Programmable Gain Amplifier
Max VCC current before configuration (AUX output on): 0.225A
Max VCC current after configuration turns off AUX output: 0.165A
Max VDD current: 0.35mA
+3V3_PGA
+VPGA
+VBIAS
PGA_BIAS
PGA_CS
PGA_SDI
PGA_SCLK
OUT_R P
OUT_R N
VCM
LMH6518
DC_FB_TRIM
Aleksa Bjelogrlic
EEVengers
Sheet: /
File: FE_Channel.kicad_sch
Title: ThunderScope
Size: A3
Date:
Rev: 5.3
KiCad E.D.A. kicad (5.1.4) -1
Id: 1/1

Clock Generator



Aleksa Bjelogrlic	
EEVengers	
Sheet: /	
File: PLL.kicad_sch	
Title: ThunderScope	
Size: A4	Date: Rev: 5.3
KiCad E.D.A. kicad (5.1.4)-1	Id: 1/1


User IO Flat Flex Connector
FPGA
Always On Clock
Status LED
Sync LVDS Transceiver
Probe Compensation
Spring Connectors
Sheet: FPGA IO Banks
File: FPGA_Bank_IO.kicad_sch.sch
Sheet: FPGA Config and Transceivers
File: FPGA_CFG_MGT.kicad_sch.sch
Sheet: FPGA Power Inputs
File: FPGA_PWR.kicad_sch.sch
Sheet: FPGA Voltage Regs
File: FPGA_VREG.kicad_sch.sch
Aleksa Bjelogrlic
EEVengers
Sheet: /
File: FPGA.kicad_sch
Title: ThunderScope
Size: A4
Date:
Rev: 5.3
KiCad E.D.A. kicad (5.1.4)-1
Id: 1/1

FPGA Configuration

A High signal on the DONE pin indicates completion of the configuration sequence. The DONE output is an open-drain output by default.

Note: DONE has an internal pull-up resistor of approximately 10 kΩ. There is no setup/hold requirement for the DONE register. These changes, along with the DonePipe register software default, eliminate the need for the DriveDONE driver-option. External 330Ω resistor circuits are not required but can be used as they have been in previous generations.

Connect INIT_B to a ≤ 4.7 kΩ pull-up resistor to V_{CCO_0} to ensure clean Low-to-High transitions.

Connect PROGRAM_B to an external ≤ 4.7 kΩ pull-up resistor to V_{CCO_0} to ensure a stable High input, and

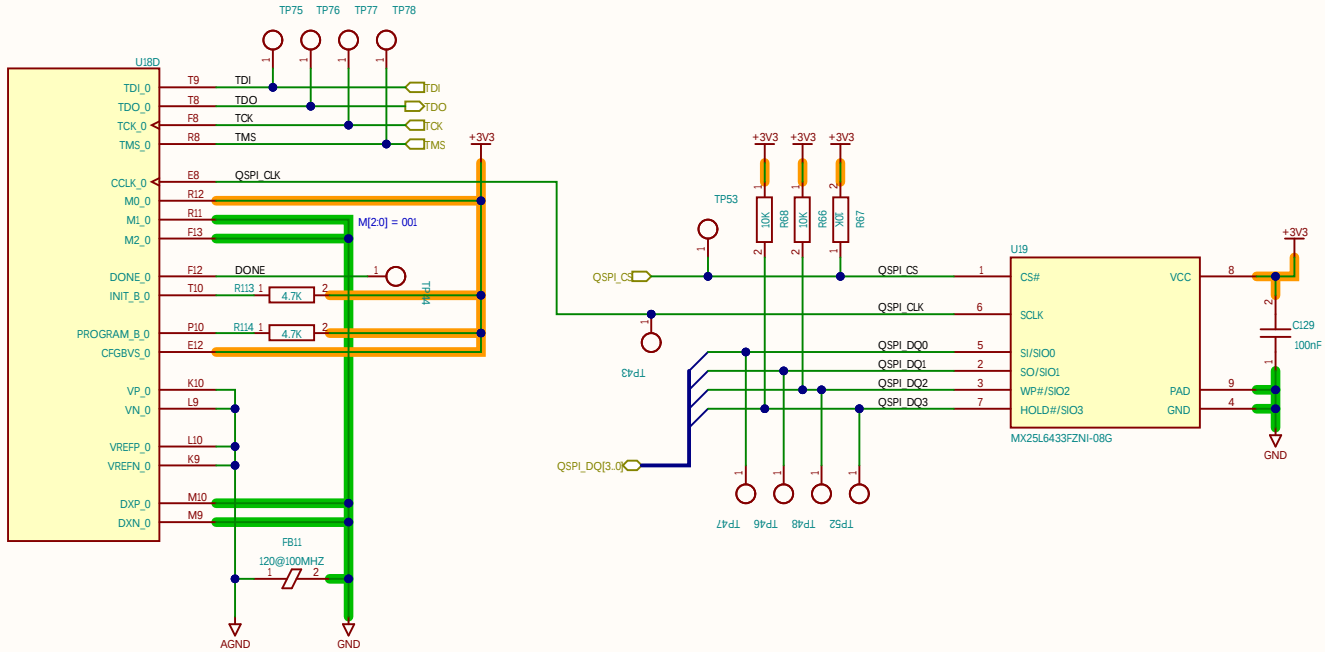
Table 2-1: **7 Series FPGA Configuration Modes**

Configuration Mode	M[2:0]	Bus Width	CCLK Direction
Master SPI	001	x1, x2, x4	Output

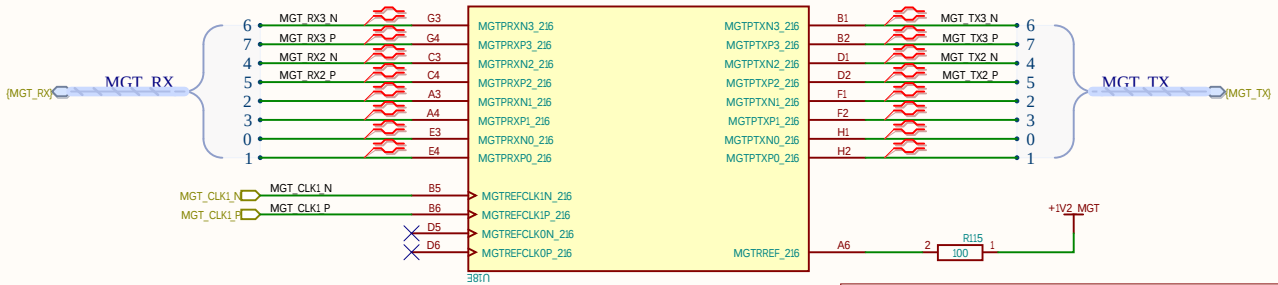
Table 2-6: **Spartan-7, Artix-7 and Kintex-7 FPGA Configuration Mode, Compatible Voltages, and CFGBVS Connection**

Configuration Mode	Banks Used	Configuration Interface I/O Voltage	HR Bank 0 V_{CCO_0}	HR Bank 14 V_{CCO_14}	HR Bank 15 V_{CCO_15}	CFGBVS
JTAG (only)	0	3.3V	3.3V	Any	Any	VCCO_0
		2.5V	2.5V	Any	Any	VCCO_0
		1.8V	1.8V	Any	Any	GND
		1.5V	1.5V	Any	Any	GND
Serial, SPI, or SelectMAP	0, 14[1]	3.3V	3.3V	3.3V	Any	VCCO_0
		2.5V	2.5V	2.5V	Any	VCCO_0
		1.8V	1.8V	1.8V	Any	GND
		1.5V	1.5V	1.5V	Any	GND
HR[2]	0, 14, 15	3.3V	3.3V	3.3V	3.3V	VCCO_0
		2.5V	2.5V	2.5V	2.5V	VCCO_0
		1.8V	1.8V	1.8V	1.8V	GND
		1.5V	1.5V	1.5V	1.5V	GND

Notes:
1. HSI/O for MapAddress or FallBack are in bank 15 but are typically only used in HR mode and not supported in HP mode.
2. HR mode is not available in the Spartan-7 family.

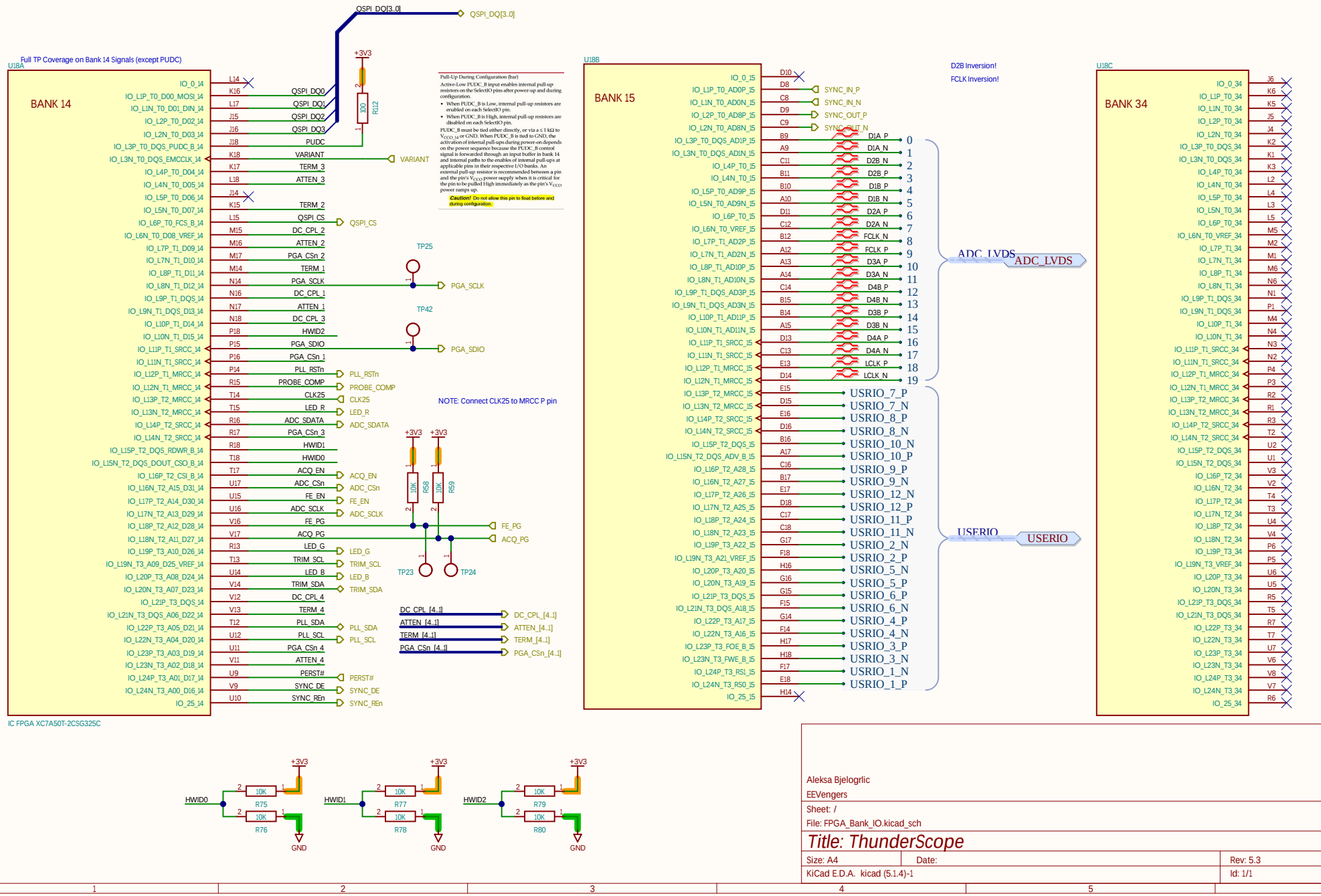


FPGA Transceivers



Aleksa Bjelogrlic		
EEVengers		
Sheet: /		
File: FPGA_CFG_MGT.kicad_sch		
Title: ThunderScope		
Size: A4	Date:	Rev: 5.3
KiCad E.D.A. kicad (5.1.4)-1		Id: 1/1

FPGA IO Banks



Aleksa Bjelogrlic
EEVengers

Sheet: /

File: FPGA_Bank_IO.kicad_sch

Title: ThunderScope

Size: A4	Date:	Rev: 5.3
KiCad E.D.A. kicad (5.1.4)-1		Id: 1/1

FPGA Power Inputs

Package	Device	V_{CCINT}						V_{CCBRAM}						V_{CCAUX}				V_{CCO} Bank 0	V_{CCO} all other Banks (per Bank)		
		680 µF	330 µF	100 µF	47 µF	4.7 µF	0.47 µF	100 µF	47 µF	4.7 µF	0.47 µF	47 µF	4.7 µF	0.47 µF	47 µF	4.7 µF	0.47 µF	47 µF	47 µF or 100 µF[3]	4.7 µF	0.47 µF
CSG325	XC7A35T XA7A35T	0	0	1	0	2	3	0	1	0	1	1	2	3	1	1	2	4			

3. One 47 µF (or 100 µF) capacitor is required for up to four V_{CCO} banks when powered by the same voltage.

Capacitor	Package Pins			Value
	MGTAVCC	MGTAVTT	GND	
Cap1		F3	F4	0.1 µF
Cap2		A2	A1	
Cap3	B4		B3	
Cap4	F5		F6	

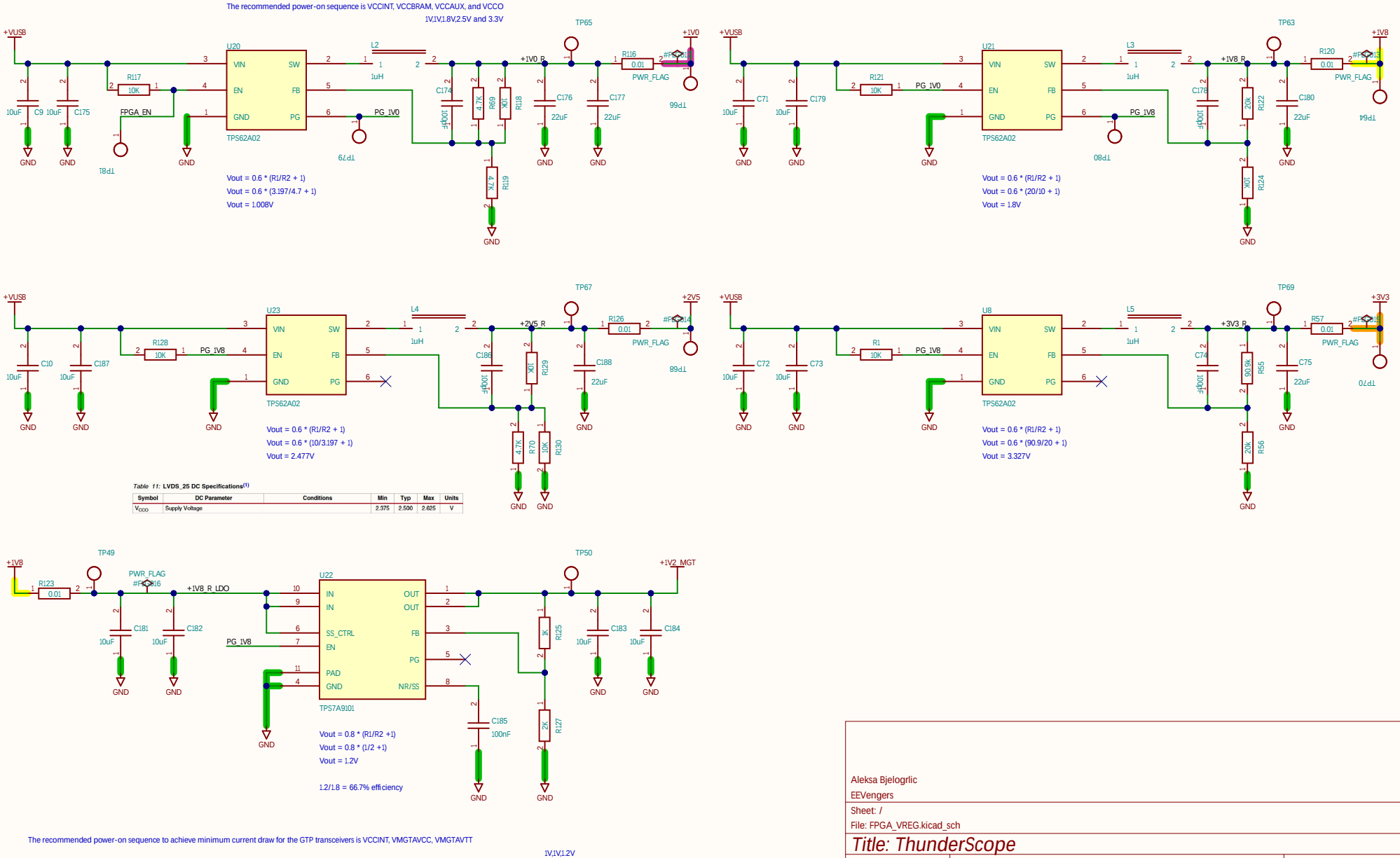
Qty/Power Supply Group		Capacitance (µF)	Tolerance	Type
MGTAVCC	MGTAVTT			
1	1	4.7	10%	Ceramic
2	2	0.1	10%	Ceramic

Aleksa Bjelogrlic
EEVengers
Sheet: /
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Title: ThunderScope

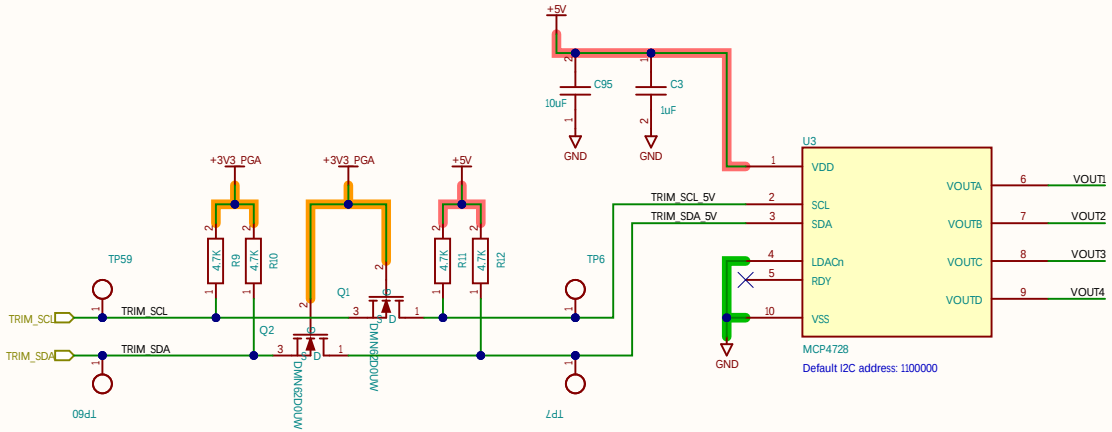
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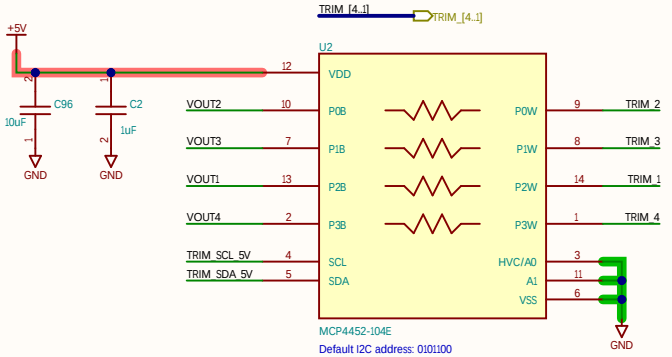
FPGA Voltage Regulators



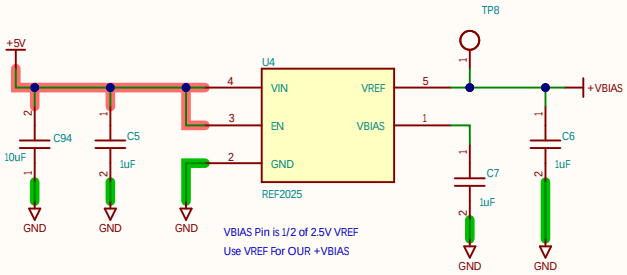
Aleksa Bjelogrlic			
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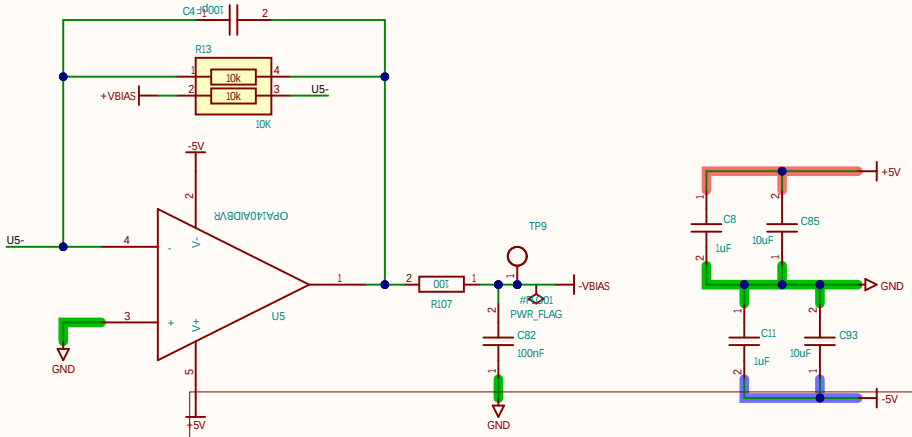
Offset Voltage Trim and User Offset Control





Bias Voltage Generation

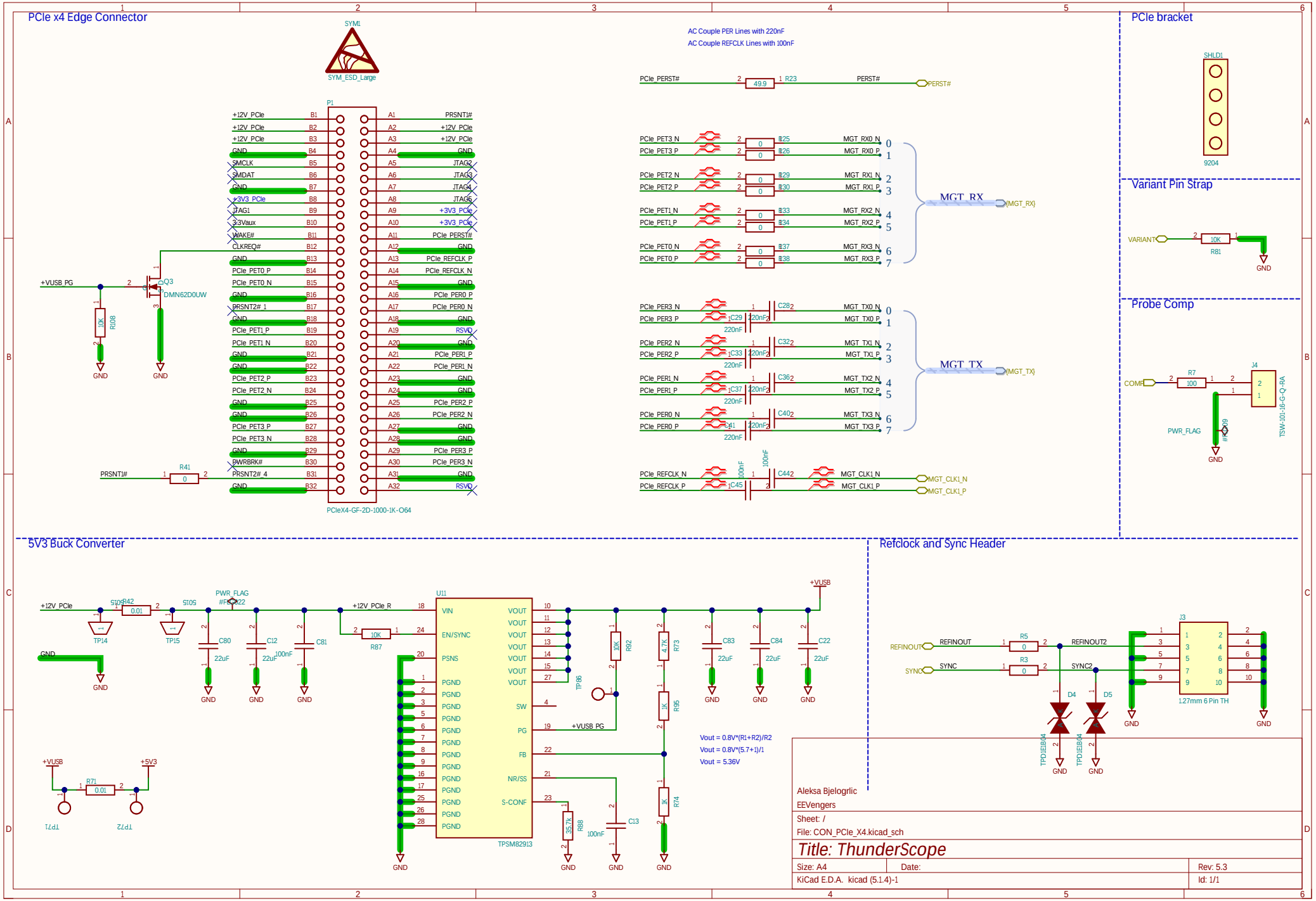




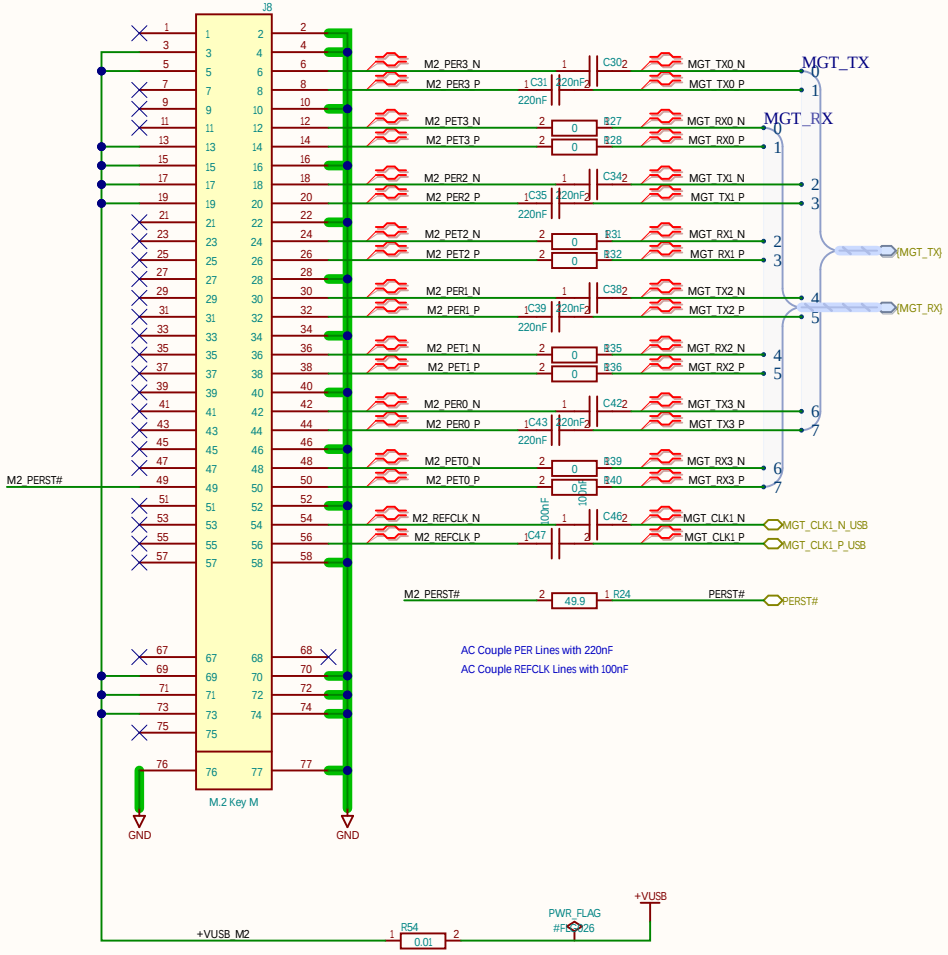
Aleksa Bjelogrlic
EEVengers
Sheet: /
File: FE.kicad_sch

Title: ThunderScope

Size: A4	Date:	Rev: 5.3
KiCad E.D.A. kicad (5.1.4)-1		Id: 1/1


PCIe x4 Edge Connector
PCIe bracket
Variant Pin Strap
Probe Comp
5V3 Buck Converter
Refclock and Sync Header
AC Couple PER Lines with 220nF
AC Couple REFCLK Lines with 100nF
SYM_ESD_Large
PCIeX4-GF-2D-1000-1K-O64
DMN62D0UW
TPSM82913
1.27mm 6 Pin TH
9204
Vout = 0.8V*(R1+R2)/R2
Vout = 0.8V*(5.7+1)/1
Vout = 5.36V
Aleksa Bjelogrlic
EEVengers
Sheet: /
File: CON_PCIe_X4.kicad_sch
Title: ThunderScope
Size: A4
Date:
Rev: 5.3
KiCad E.D.A. kicad (5.1.4)-1
Id: 1/1

M.2 Key M Connector - Custom Pinout



AC Couple PER Lines with 220nF

AC Couple REFCLK Lines with 100nF

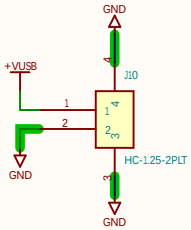
NOTE: The TB/USB4 adaptor must be modified to give us VUSB instead of 3V3

Interposer Standoffs



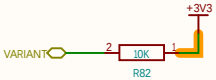
Probe Comp

Fan Connector

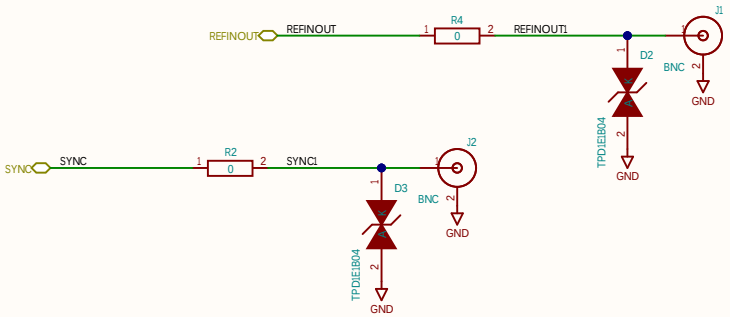


Ground Lug

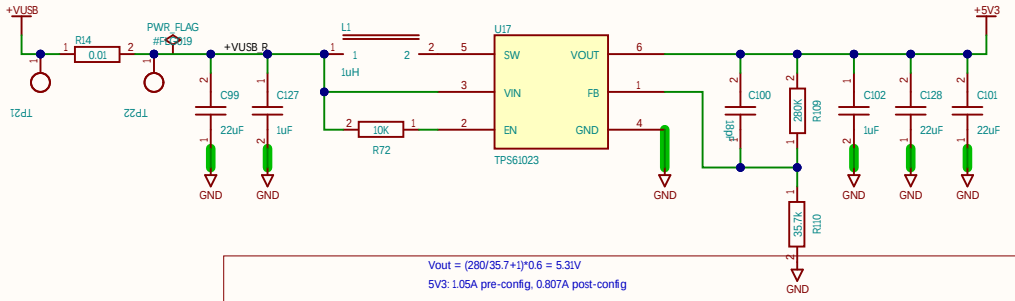
Variant Pin Strap



Refclock and Sync BNCs



5V3 Boost Converter



Vout = (280/35.7+1)*0.6 = 5.31V

5V3: 1.05A pre-config, 0.807A post-config

Aleksa Bjelogrlic		
EEVengers		
Sheet: /		
File: M2_KEY_M.kicad_sch		
Title: ThunderScope		
Size: A4	Date:	Rev: 5.3
KiCad E.D.A. kicad (5.1.4)-1		Id: 1/1